Transfer: circuits and induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply circuit relationships and explain an induction observation.

SUCCESS CRITERIA - I can...

1. Use current and voltage rules in a new circuit.
2. Account for electrical power and energy.
3. Explain induction through changing flux.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A motor is **62%** efficient and the following generator is **71%** efficient. **Find the combined useful fraction** in sequence.

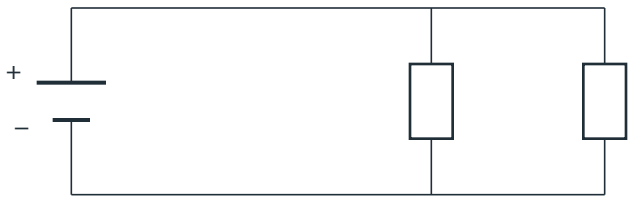
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $V = IR$; $P = IV$; $E = P\Delta t$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$. Use stated facts about the circuit or coil. Ideal battery and wires unless stated otherwise.

Worked reasoning: A **12 V** supply drives a **6 ohm** resistor: current **2 A** and power **24 W**. A complete explanation checks both the charge-flow rate and the energy-transfer rate.

Your turn: 1. **Draw the circuit** and an energy-flow arrow.
2. **Label what an ammeter** and voltmeter would measure and where each connects.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Sketch two resistors in parallel across a battery and **label the equal-voltage branches**.

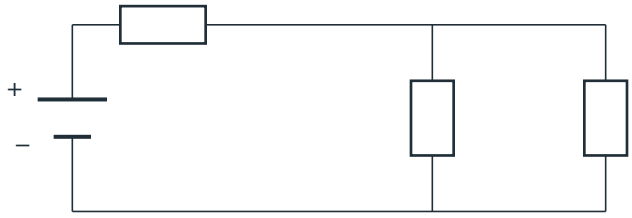


Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A **12 V** battery powers a **4 Ω** resistor in series with two parallel **8 Ω** resistors.
1. **Label the network**.
2. **Calculate total current**, each branch current and total input power.



Marzano check - circle: ① ② ③

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05 Explain your reasoning

A magnet is held motionless near a fixed coil, then moved into it. **Explain the change** in the meter response without saying the magnet supplies unlimited energy.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Measured (I, V) pairs are (**0.10**, **4**), (**0.20**, **8**), (**0.30**, **12**) in A, V. **Find resistance** and **explain how the measurements support** treating the resistance as constant.
Draw the stated circuit and **label quantities relevant** to the question.

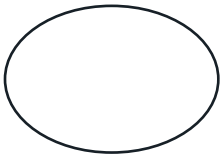
Draw the complete circuit and label shared nodes.

Marzano check - circle: ① ② ③

07 Exit check

A **50**-turn coil has flux per turn change by **0.004 Wb** in **0.20 s**. **Find average emf magnitude**.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **12 V** source feeds a **2 ohm** series resistor and a parallel pair of **4 ohm** resistors. **Find total power** and power in each resistor; verify the energy-rate sum.



Marzano check - circle: ① ② ③